Flavio's Comments — ID → Anchor → Text Mapping

Reference for `flavio_comments_review.md`. Total comments: **66**. The `#` is Word's internal comment ID (used in the review file), not the visual order of bubbles. "Anchor" = the document text the comment is attached to.

#2

Anchored to: *Infra*

Flavio: For a paper, I would use a less technical title. This one es very correct, but hard to understand. It may also depend on how specific the journal is, but often a more broadly understandable title helps.E.g. Healthy, but not pathological/clinical, aging alters infra-slow fluctuations of sleep sigma/spindle power

#4

Anchored to: *Abstract*

Flavio: Already on a great level in my opinion. My main suggestion is, as for the title, write it to be understandable for a more broader audience. My specific notes:- First two sentences are too long and a bit repetitive.I would not state that almost none has looked at ISFS in old instead “just” state what is unknown.The technical part is great. “and measuring the peak frequency, bandwidth, and area under the spectral peak (AUC) of the sigma-envelope spectrum at every channel” is too specific in my opinionYour results are a bit too technical to me. E.g. Instead of “ Peak frequency was faster in both older groups than in young adults (p = 0.0026), with a parallel, non-significant tendency toward broader bandwidth (p = 0.061). “ Slow waves become smaller/faster/more high frequency with age, compared to young ones. -> in better language, but I hope you get my point. End with the main findings impact/outlook etc. Not on something that did not work.

#6

Anchored to: *Non*

Flavio: Good paragraph on explaining what and what for spindles are. Not sure if the best opening paragraph for a paper, maybe something that makes the reader more curious or is closer to the paper’s quetsion.

#7

Anchored to: *most*

Flavio: A large part?

#8

Anchored to: *Spindles*

Flavio: I had to re-read it to understand what the main takeaway point is here. I felt you ment: spindle trains appear in a specific, slow rhythm of 0.02hz (more my style would be “every 50s”). I would state this more explicitly and then add what this technically means (envelope etc) and why this is the case (LC, noradrenaline)

#9

Anchored to: *Dimitriades et al. (2024) subsequently characterized the ISFS across development and in young adults, quantifying per channel the peak frequency, spectral width, and area under the spectral peak of the sigma envelope, and linking these to markers of arousal and memory reactivation[11]. The same infra-slow grouping of spindles tracks memory consolidation in humans[12].*

Flavio: I don’t understand the difference between the two sentences.

#10

Anchored to: *Healthy*

Flavio: Nice!

#11

Anchored to: *Yet*

Flavio: great

#12

Anchored to: *departs from healthy aging or instead mirrors it*

Flavio: I did not get this [redacted].whether ISFS in MCI is comparable or different from ISFS in healthy elderly?

#15

Anchored to: *Participants*

Flavio: Clear and good

#17

Anchored to: *recordings*

Flavio: I would give it a bit more structure.People came to the sleep lab, EEG was applied, they slept for XX hours. The goal was to extract participants ISFS.To get clean ISFS we preprocessed the data …. Thow away channelsN2 bouts: We sleep-scored (cite AASM) the whole night and removed all not N2 epochs. Further we only kept 300s bouts of N2 without bad epochs…ISFS extraction...

#18

Anchored to: *An overview of the recorded sleep across the three groups, including whole-night hypnospectrograms, the distribution of sleep stages, and N2 bout properties, is shown in Figure 1.*

Flavio: For me this is already a results

#25

Anchored to: *For each clean bout, the envelope was mean-subtracted to remove the DC component and Fourier-transformed; the squared-magnitude spectrum was interpolated onto a common frequency axis and normalized by its own mean.*

Flavio: This I don't understand. You subtract the mean and then normalize again by the mean?"to remove the DC component and Fourier-transformed" sounds not like a sentence and I don't understand. What is transfomred? What is DC?

#26

Anchored to: *The normalized bout spectra were averaged within a channel, and a baseline estimated from 0.06–0.10 Hz, above the ISFS band, was subtracted to flatten the spectral floor.*

Flavio: This is a nice hack. I had to read it twice to understand. So maybe you first say what the issue is and how you tackled it with the subtraction of the higher baseline.

#27

Anchored to: *A Gaussian, a·exp(–((f–b)/c)²), was fitted to the baseline-corrected mean spectrum by nonlinear least squares. A fit was accepted only if its peak amplitude exceeded the detection threshold (1.5 times the standard deviation of the spectrum), its peak frequency fell within the ISFS band 0.0075–0.04 Hz, and the numerically integrated area under the fitted peak was positive, which rejects degenerate …*

Flavio: Also here, I would first state what you want to to do.Like.To capture ISFS, we estimated peak frequence bandwith and AUC of the power in the ISFS band. If peak amplitude of the bout was outside (XXX) we rejected the bout. To receive peak amplitude we fitted a Gaussian kernel...

#28

Anchored to: *A Gaussian, a·exp(–((f–b)/c)²), was fitted to the baseline-corrected mean spectrum by nonlinear least squares. A fit was accepted only if its peak amplitude exceeded the detection threshold (1.5 times the standard deviation of the spectrum), its peak frequency fell within the ISFS band 0.0075–0.04 Hz, and the numerically integrated area under the fitted peak was positive, which rejects degenerate …*

Flavio: And refer to the figure. The figure helps a lot

#30

Anchored to: *Region*

Flavio: Nice. clear, easy to understand and still precise

#32

Anchored to: *. ISFS concept, feature extraction, and the central-parietal ROI.*

Flavio: I would choose something simpler. LikeCharacterization of infra-slow fluctuation of sigma power

#33

Anchored to: *The green box marks the window expanded in B)*

Flavio: Somehow that you refer to B triggered me ■. Probably a me problem

#35

Anchored to: *whole-scalp, topographic*

Flavio: I don’t understand the difference without reading the rest.

#36

Anchored to: *scalp*

Flavio: For me it is not whole scalp. More like “main ISFS”. Or Overall ISFS.

#37

Anchored to: *when all groups were*

Flavio: Where they not always normal?It is better to write it not in hypothetical. As you only did a few group comparison it is easier to say as they are.

#38

Anchored to: *cluster-based permutation tests*

Flavio: Love it and well described

#39

Anchored to: *Throughout, the group mean displayed on a topography is the mean of the per-subject means, computed independently of the plotted image and matching the violin-plot statistic; missing channels in a topographic image were filled by spatial neighbor imputation for display only, never for the group statistics. Correlations between ISFS parameters and MoCA were assessed within the pooled older-adult an …*

Flavio: This is a bit confusing.Make an uncopuled statement about correlations.Then I understand the start, but this not:"computed independently of the plotted image and matching the violin-plot statistic”. Last, I would restructure the middle: missing electrode values were interpolated for visualization purposes only.

#40

Anchored to: *. Demographic variables were compared between the two older groups (age and MoCA) under the same normality-gated scheme: a Welch t-test when both groups passed Shapiro–Wilk and a Mann–Whitney U test otherwise. Group differences in sleep-stage proportions were tested with the same three-group omnibus and post-hoc scheme as the whole-scalp ISFS parameters.*

Flavio: This is okey, but could be way simpler if you state the result here.Age and MoCA did not differ between healthy eldery and MCI (normal distribtution, Welch t-test). Sleep-staging did (not) differ between groups (young, healthy elderly, MCI), tested with an omnibus ANOVA? And ASDF posthoc comparisons.

#44

Anchored to: *Cohort*

Flavio: I prefer that the title states the main results instead of describe the category.

#46

Anchored to: *ISFS*

Flavio: For me it takes a bit too long until you get to “interesting” results (4.3). Personally I would write 4.1 and 4.2 in methods and summarize it with two sentence in result. You can absolutely describe it here, then I would make it easier to crasp the main thing.

#48

Anchored to: *parameters*

Flavio: I would name the parameter

#49

Anchored to: *parameters*

Flavio: SIFS frequency changes with ageing

#50

Anchored to: *age*

Flavio: I would start with a sentence of what you were interested in finding/ comparing etc. The sentence right now is too technical.

#51

Anchored to: *age*

Flavio: We wondered ….For this we compare peak frequency, bandwidth and AUC between the groups X,Y,Z

#54

Anchored to: *On every parameter, the elderly and MCI groups were statistically indistinguishable.*

Flavio: I can’t really say why, but this feels weird to me. I would rather say something like: Neither peak frequency, bandwidth nor strength differed significantly between elderly and MCI patients.

#55

Anchored to: *kernel density*

Flavio: I think it is too technical. half-violin estimates the distribution.

#56

Anchored to: *Peak frequency was higher in both older groups than in young adults (young < elderly = MCI; one-way ANOVA p = 0.0026, post-hoc young vs elderly p = 0.013, young vs MCI p = 0.005). Bandwidth trended in the same direction but did not reach significance (one-way ANOVA p = 0.061; no post-hoc tests).*

Flavio: I think you should start with this as the figure description.And I would focus more on the effect you have then all the ones you did not turn out to be significant.

#58

Anchored to: *ISFS strength is focally reduced over central-parietal cortex*

Flavio: GREAT!maybe add for which group

#59

Anchored to: *Although*

Flavio: Again: GREAT

#60

Anchored to: *Although*

Flavio: I think most result paragraph should be written like that. First in simple words what you find, then technical specifics.

#62

Anchored to: *Young adults show higher AUC over the central-parietal electrodes, and the hotspot flattens and spreads in aging and MCI*

Flavio: Start with this!

#64

Anchored to: *To quantify the central-parietal effect within the pre-defined region of interest, each subject's per-channel AUC was first normalized by that subject's whole-scalp mean and then averaged over the ROI electrodes (Figure 5).*

Flavio: methods

#65

Anchored to: *The*

Flavio: Maybe start with what did you ask.

#66

Anchored to: *The*

Flavio: Again, start with this

#67

Anchored to: *The*

Flavio: And be more explicit. ISFS strength descriptively declined from young to older participants, however the group comparison remained not significant (one-way ANOVA.)

#70

Anchored to: *The same cluster-based permutation procedure was applied to the peak-frequency and bandwidth maps (Figure S1).*

Flavio: Technical statistics. What and why did you want to compare.E.g.The comparison of young adults, elderly,… did not significiantly differ in there ISFS frequency and bandwidth (M = ,…. Cluster-bades perm. stats,...)

#71

Anchored to: *The same cluster-based permutation procedure was applied to the peak-frequency and bandwidth maps (Figure S1).*

Flavio: If someone wants to know whether its peak-frequency or which cluster stas they can have a look at the mehtods

#73

Anchored to: *Within the pooled older-adult and MCI sample*

Flavio: Also technical description.We did not find a relationship between the participants' ISFS expression and cognitive capacity (MoCA score). Specifically, neither ISFS frequency, bandwidth nor strength correlated with the MoCA score (all r<xx)

#77

Anchored to: *Discussion*

Flavio: You worte a lot of how you interpret your results. Generally it is great that you thought about it that deeply.Now there are two issues:It is a bit hard to understandYuval's style is often more descriptive than interpretation.I suggest that you make shorter and simpler paragraphs. Maybe you can try to reduce each paragraph to one conclusion.

#78

Anchored to: *Applying*

Flavio: Already great

#81

Anchored to: *groups*

Flavio: I think you need one sentence that recaps what you wanted to examine and how you did this.E.g. We set out to…Therefore we compared the ISFS across three groups, namely ….

#94

Anchored to: *adults*

Flavio: With a few premise (like what is ISFS), most of the discussion should be understand by your friends.Like instead of "Young adults showed a pronounced spectral-area hotspot that flattened with age (cluster p = 0.023). " write on the level Compared to young adults, ISFS is less focal and weakens with age.

#96

Anchored to: *A parallel tendency toward broader spectral width was present in the same direction but did not reach significance; this effect was borderline and sensitive to which subjects were included, having reached significance in an earlier subject set, and is best read as a trend rather than an established effect*

Flavio: You cannot add new results here (like how it was in earlier sets). If it is important for the discussion you need to report it in the results.However, I think you should not make a bit thing of the trend result. Just state as is

#98

Anchored to: *Most consequentially, patients with MCI were statistically indistinguishable from healthy older adults on every ISFS measure, and no ISFS parameter tracked the MoCA.*

Flavio: If this is your most important results, you should check it statistically. The bayesian framework gives you an easy way to check how likely it is that the two old groups are similar vs different.

#104

Anchored to: *Both directional changes*

Flavio: Name them. Repetition is good.

#105

Anchored to: *The flattening of that hotspot together with an acceleration of the rhythm suggests that the roughly 0.02 Hz clock pacing spindle trains becomes both faster and less spatially concentrated, so that its energy is no longer focused where fast-spindle generators are densest. A similar loss of spatial structure was visible, though it did not reach significance, in the peak-frequency map: the frontal e …*

Flavio: Could be simplified, as it is mainly a repetition of the findings.

#106

Anchored to: *A focal reduction in ISFS strength therefore need not translate proportionally into disorganized spindle output, and the AUC effect is more safely interpreted as a sensitive index of how the rhythm is distributed than as direct evidence of degraded spindle function.*

Flavio: This is don't undrestand

#107

Anchored to: *The*

Flavio: Great paragraph in my opinion. Could sometimes be written a bit more simple, but generally good

#108

Anchored to: *plausible substrate*

Flavio: Very LLM

#109

Anchored to: *negative result*

Flavio: Right now you don't have a negative results, see my bayesian comment

#110

Anchored to: *The*

Flavio: Very good paragraph.

#112

Anchored to: *not measurably accelerate,*

Flavio: Or ISFS changes are predictive before MCI is detectable. Meaning if it relates to the factor that is attacking cognition, only later cognition is impaired.

#113

Anchored to: *, and its insensitivity to MCI is itself informative about what the rhythm indexes*

Flavio: I don't think I understand.it is informative about what index?

#114

Anchored to: *They found the amplitude of the rhythm selectively reduced in AD, with its frequency and bandwidth preserved; the amplitude reduction tracked the plasma amyloid ratio, and bandwidth tracked markers of neurodegeneration and poorer memory retention. Their amplitude result parallels our central-parietal AUC cluster, since both implicate a weakening of the rhythm's strength over central regions, altho …*

Flavio: This is technically very correct.For me it would be easiert for the reader, if you just call both ISFS strength and maybe explain the amplitude in a bracket( ).

#115

Anchored to: *Because their study appeared after the present analysis was complete, the two are best read as independent and largely convergent evidence that the infra-slow sigma rhythm is sensitive to both aging* *and neurodegeneration.*

Flavio: Also too much LLM for me. A) you don’t show that ISFS is sensitive to neurodegeneration. B) not convergent evidence C) what appeared when is confusing.I would just end with both studies show age effect, whereas relation of ISFS with AD but not MCI has been found.

#116

Anchored to: *Several*

Flavio: Too many limitations. I would choose 3 and discuss them properly. First describe, add why you couldn’t change this and maybe was in the future can be done or what’s the issue with it is.

#118

Anchored to: *Future*

Flavio: great

#119

Anchored to: *It would also be informative to examine how the ISFS couples to other infra-slow rhythms of sleep, including the slow oscillation and infra-slow hemodynamic fluctuations,*

Flavio: Why not.But there has been a lot of rodent work to make sure ISFS is meaningful.Okay now I understand. You mean coupling ISFS to SW peaks for example. Then I think you need to explain a bit more. Why do you think this could be interesting or exist.

#120

Anchored to: *pursue the rodent locus-coeruleus account translationally, given that the mechanistic origin of the human rhythm remains inferred rather than measured.*

Flavio: This I don’t understand.